
Latency as a regularizer for reward shaping in reinforcement learning

A modern view on M. Robin’s reduction

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Abstract

We revisit a classical reduction, due to M. Robin [5, 6], of impulse-control problems with execution delay to no-delay problems on the same state space, in modern reinforcement-learning language. Read as an identity between expected-return functionals, the reduction states that an agent receiving reward $r(a, s_{(t+\tau)-})$ at decision time t has the same value, optimal policy, and expected return as an agent receiving a smoothed reward at decision time. The state space and the underlying state dynamics are unchanged — the delay of the intervention is reflected on the effective reward only. This rewrites a question on value functions as a question on rewards and supplies a principled prescription for soft-margin reward shaping in delayed-action reinforcement learning. Our justification includes a state-dependent random delay, allowing latency modelling to reflect clustering effects of latency.

1 Introduction

Action-delivery delay is a feature of many controlled systems: decision-making on stale information in remote operation, frame-time variability in interactive software where the agent’s commit lands a few frames later. A consistent qualitative observation across these settings is that the optimal policy under increased delay is smoother than its no-delay counterpart — less reactive to high-frequency state fluctuations, shifting from sharp to conservative behaviour, with effective dead zones around action boundaries (unless the reward specifically penalises non-sparse actions). The agent acquires a form of regularization as a function of the environment’s response time.

In essence, this regularization is the 1976 result of M. Robin [5, 6]: a delayed-action impulse-control problem reduces to a no-delay problem on the same state space whose intervention operator is integrated against the inter-decision transition kernel. In reinforcement-learning language, delayed-action evaluation is equivalent to undelayed evaluation on a state-space-smoothed reward.

Practitioners in the ML community routinely replace the sparse indicator rewards used to specify a task with hand-engineered soft margins, both to obtain gradient signal during training and to reduce the variance of TD-target updates. The choice of margin is heuristic. The reduction specifies the unique soft margin that preserves the lagged optimum: it is the convolution of the original reward with the inter-decision transition kernel. The same construction therefore both supplies a structural justification for the variance-reduced training signal practitioners write down on intuition, and characterizes the optimum that signal targets.

We restate Robin’s reduction in elementary RL terms, develop its content as a prescription for soft-margin reward shaping, and extend it to a state-dependent random delay $\mu(\cdot \mid s, a)$ that allows modelling clustering effects in latency. A numerical illustration on a one-dimensional environment with a multi-modal reward landscape exposes the regularization mechanism directly: the latency-induced Gaussian smoothing kernel selectively suppresses fragile narrow optima while preserving robust wide ones, producing a discontinuous bifurcation of the optimal aim point at an analytically predicted latency ε_c . The smoothed-reward training signal additionally has lower reward variance, hence lower TD variance.

2 Setup

Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{P})$ carry an $\mathcal{S}$ -valued state process $(s_t)_{t \geq 0}$, with $\mathcal{S}$ a continuous state space and $\mathcal{A}$ a measurable action space. Decisions are taken sequentially at times $T_1 = 0 < T_2 < \dots$ according to a policy $\pi: \mathcal{S} \rightarrow \mathcal{P}(\mathcal{A})$. At each decision epoch T_n ,

$$a_n \sim \pi(\cdot \mid s_{T_n}), \quad \tau_n \sim \mu(\cdot \mid s_{T_n}, a_n), \quad T_{n+1} = T_n + \tau_n, \quad (1)$$

with a_n measurable with respect to $\mathcal{F}_{T_n}$. Action a_n is dispatched at T_n and lands at $T_n + \tau_n$; during the lag $[T_n, T_{n+1})$ the state (s_t) continues to evolve. A reward is a measurable function $r: \mathcal{A} \times \mathcal{S} \rightarrow \mathbb{R}$, evaluated for action a_n at the state immediately before delivery, $s_{(T_n + \tau_n)^-}$. We make the reward an explicit argument of the return functional and compare, on a common policy class, the undelayed return

$$G(\pi; r) = \mathbb{E} \left[\sum_{n \geq 1} e^{-\rho T_n} r(a_n, s_{T_n}) \right], \quad \rho > 0,$$

and its lagged counterpart

$$\tilde{G}(\pi; r) = \mathbb{E} \left[\sum_{n \geq 1} e^{-\rho T_n} r(a_n, s_{(T_n + \tau_n)^-}) \right].$$

We will use $G(\pi; r)$ and $\tilde{G}(\pi; r)$ throughout.

Assumption A (Measurability of the kernels). *The conditional law of $s_{(T_n + \tau_n)^-}$ given $\mathcal{F}_{T_n}$, a_n , and τ_n admits a density $f_\tau(\cdot \mid s_{T_n}; a_n)$ with respect to a fixed reference measure on $\mathcal{S}$, and the maps $(\tau, s, s', a) \mapsto f_\tau(s' \mid s; a)$ and $(s, a) \mapsto \mu(\cdot \mid s, a)$ are measurable kernels.*

For the dynamic-programming form of §3.2 we additionally assume that (s_t) is Markov between decisions; this is the $m = 1$ Robin setup of [6, Ch. VI].

Assumption B (Markov state). *The state process (s_t) is Markov in the no-pending phase: conditional on s_t , the future $(s_u)_{u \geq t}$ is independent of $\mathcal{F}_t$.*

We will assume that the reward r is bounded.

3 Latency and effective reward smoothing

3.1 Equivalence of cost functionals

Theorem 3.1. *Under Assumption A, for every $\mathcal{F}_{T_n}$ -measurable policy π ,*

$$\tilde{G}(\pi; r) = G(\pi; \tilde{r}), \quad \tilde{r}(a, s) := \int_{\mathcal{S}} r(a, s') \mathbb{E}_{\tau|s,a} [f_\tau(s' \mid s; a)] ds'.$$

In particular, the lagged and smoothed-reward problems share the same expected return on every policy, the same optimal value, and any policy optimal in one is optimal in the other.

Proof. By definition of $\tilde{G}$, Fubini for the sum and the expectation, the tower property conditioning on $(\mathcal{F}_{T_n}, a_n, \tau_n)$, Assumption A for the density of $s_{(T_n + \tau_n)^-}$, and marginalization of τ_n against $\mu(\cdot \mid s_{T_n}, a_n)$:

$$\begin{aligned} \tilde{G}(\pi; r) &= \mathbb{E} \left[\sum_{n \geq 1} e^{-\rho T_n} r(a_n, s_{(T_n + \tau_n)^-}) \right] \\ &= \sum_{n \geq 1} \mathbb{E} \left[e^{-\rho T_n} \mathbb{E} [r(a_n, s_{(T_n + \tau_n)^-}) \mid \mathcal{F}_{T_n}, a_n, \tau_n] \right] \\ &= \sum_{n \geq 1} \mathbb{E} \left[e^{-\rho T_n} \mathbb{E}_{\tau_n} \left[\int_{\mathcal{S}} r(a_n, s') f_{\tau_n}(s' \mid s_{T_n}; a_n) ds' \right] \right] \\ &= \sum_{n \geq 1} \mathbb{E} \left[e^{-\rho T_n} \int_{\mathcal{S}} r(a_n, s') \mathbb{E}_{\tau_n} [f_{\tau_n}(s' \mid s_{T_n}; a_n)] ds' \right] \\ &= \sum_{n \geq 1} \mathbb{E} \left[e^{-\rho T_n} \tilde{r}(a_n, s_{T_n}) \right] = G(\pi; \tilde{r}), \end{aligned}$$

where the last line recognizes the definition of $\tilde{r}$. The equality holds on a common policy class, so any policy optimal in one is optimal in the other. $\square$

3.2 Dynamic programming with delay

Under Assumptions A and B, the equivalence of §3.1 yields a Bellman equation for the lagged problem on $\mathcal{S}$ alone, in terms of the smoothed reward of Theorem 3.1 and a discounted inter-decision kernel. Let

$$V(s) := \sup_{\pi} \tilde{G}(\pi; r \mid s)$$

denote the lagged value function for the discounted return started from $s_{T_1} = s$ in the no-pending phase, and define the discounted inter-decision kernel

$$\tilde{K}_{\mu}^{\rho}(s' \mid s; a) := \int_0^{\infty} e^{-\rho\tau} f_{\tau}(s' \mid s; a) \mu(d\tau \mid s, a).$$

Theorem 3.2 (Bellman equation with delay). *If $\kappa := \sup_{s,a} \int_0^{\infty} e^{-\rho\tau} \mu(d\tau \mid s, a) < 1$, then V is the unique bounded measurable solution of*

$$V(s) = \sup_{a \in \mathcal{A}} \left\{ \tilde{r}(a, s) + \int_{\mathcal{S}} V(s') \tilde{K}_{\mu}^{\rho}(s' \mid s; a) ds' \right\}, \quad (2)$$

with $\tilde{r}$ the smoothed reward of Theorem 3.1.

Equation (2) is the Bellman equation of an undelayed Markov decision problem on $\mathcal{S}$ with reward $\tilde{r}$ and transition kernel $\tilde{K}_{\mu}^{\rho}$. The reward $\tilde{r}$ is the same object as in Theorem 3.1; the discount $e^{-\rho\tau}$ enters only through the kernel, accounting for the time-shift between consecutive decision epochs.

4 Applications to reward-shaping prescription

The reduction converts a question about the value function into a question about the reward: training an agent on $\tilde{r}$ with rewards undelayed (though the state effect remains delayed) recovers the policy that is optimal in the lagged environment with the original reward r that was coming in after a lag. Reinforcement-learning practitioners routinely replace sparse indicator rewards with hand-engineered soft margins to obtain gradient signal during training; the choice of margin is heuristic. Theorem 3.1 specifies the unique choice that recovers the lagged optimum: $\tilde{r}$ is the convolution of r with the inter-decision transition kernel. The next subsections provide closed-form rewards in two generic situations: constant and exponential delays.

4.1 Gaussian diffusion in state, constant delay

Consider the simplest setting: a one-dimensional Brownian state with diffusion σ during the lag and deterministic delay τ_0 . The inter-decision kernel is

$$f_{\tau_0}(s' \mid s) = \frac{1}{\sqrt{2\pi\sigma^2\tau_0}} \exp\left(-\frac{(s' - s)^2}{2\sigma^2\tau_0}\right),$$

and the smoothed reward $\tilde{r}_{\tau_0} = r * f_{\tau_0}$. In the case of two generic rewards, this yields:

Example 4.1 (Indicator becomes coverage probability). *For $r(s) = \mathbb{1}\{|s| < w/2\}$,*

$$\tilde{r}_{\tau_0}(s) = \mathbb{P}(|s + \sigma\sqrt{\tau_0} Z| < w/2 \mid s) = \Phi\left(\frac{w/2 - s}{\sigma\sqrt{\tau_0}}\right) - \Phi\left(\frac{-w/2 - s}{\sigma\sqrt{\tau_0}}\right),$$

the coverage probability of the original indicator under the inter-decision kernel.

Example 4.2 (Gaussian target becomes a wider Gaussian). *For $r(s) = \exp(-s^2/(2\ell^2))$,*

$$\tilde{r}_{\tau_0}(s) = \frac{\ell}{\sqrt{\ell^2 + \sigma^2\tau_0}} \exp\left(-\frac{s^2}{2(\ell^2 + \sigma^2\tau_0)}\right);$$

the shape is preserved with effective width $\sqrt{\ell^2 + \sigma^2\tau_0}$.

Both examples extend directly to a multivariate state $s \in \mathbb{R}^d$ by replacing the scalar variance $\sigma^2\tau_0$ with a covariance $\sigma\sigma^{\top}\tau_0$ in the kernel: the convolution stays a Gaussian heat-kernel smoothing of r . Whenever r has a characteristic length scale ℓ , the smoothing makes $\tilde{r}$ Lipschitz on the scale $\sigma\sqrt{\tau_0}$ once $\sigma\sqrt{\tau_0} \gtrsim \ell$.

4.2 Random delay in the Brownian-state setting

We stay in the setting of §4.1 (one-dimensional Brownian state with diffusion σ during the lag) and now allow the delay τ to be random. By Theorem 3.1, the inter-decision kernel becomes the τ -mixture of Gaussian kernels,

$$K_\mu(s' | s) = \int_0^\infty \frac{1}{\sqrt{2\pi\sigma^2\tau}} e^{-(s'-s)^2/(2\sigma^2\tau)} \mu(d\tau | s),$$

and the smoothed reward $\tilde{r}(s) = \int r(s') K_\mu(s' | s) ds'$. We illustrate two models for μ and, in each, evaluate the smoothing on the indicator and Gaussian rewards of Examples 4.1–4.2.

State independent model: Constant-rate exponential delay. With $\tau \sim \text{Exp}(\lambda)$ independent of s , the mixture integral has a closed form,

$$K_\mu(s' | s) = \int_0^\infty \frac{1}{\sqrt{2\pi\sigma^2\tau}} e^{-(s'-s)^2/(2\sigma^2\tau)} \lambda e^{-\lambda\tau} d\tau = \frac{1}{2\beta} e^{-|s'-s|/\beta}, \quad \beta := \frac{\sigma}{\sqrt{2\lambda}}, \quad (3)$$

a symmetric Laplace kernel of scale β . The Gaussian heat kernel of §4.1 is replaced by a heavier-tailed $e^{-|\Delta|/\beta}$ profile.

Convolving (3) with the rewards of Examples 4.1–4.2 gives:

Example 4.3 (Indicator $r(s) = \mathbb{1}\{|s| < w/2\}$). *Direct integration yields*

$$\tilde{r}(s) = \begin{cases} 1 - \frac{1}{2} \left(e^{-(w/2+s)/\beta} + e^{-(w/2-s)/\beta} \right), & |s| \leq w/2, \\ \frac{1}{2} (1 - e^{-w/\beta}) e^{-(|s|-w/2)/\beta}, & |s| > w/2. \end{cases} \quad (4)$$

The Gaussian-CDF margin of Example 4.1 is replaced by an exponentially-decaying margin of characteristic scale β outside the slab.

Example 4.4 (Gaussian target $r(s) = \exp(-s^2/(2\ell^2))$). *The convolution admits the closed form*

$$\tilde{r}(s) = \frac{\ell\sqrt{2\pi}}{2\beta} e^{\ell^2/(2\beta^2)} \left[e^{-s/\beta} \Phi\left(\frac{s}{\ell} - \frac{\ell}{\beta}\right) + e^{s/\beta} \Phi\left(-\frac{s}{\ell} - \frac{\ell}{\beta}\right) \right], \quad (5)$$

recovering the wider Gaussian of Example 4.2 in the regime $\beta \ll \ell$ and decaying at the Laplace rate $1/\beta$ in the tails $|s| \gg \ell$.

State dependent model: Hawkes-modulated rate. We now let the rate be a state coordinate driven by latency-spike events. Let N^* be an exogenous Hawkes counting process whose jumps record latency spikes, with exponential self-excitation kernel and conditional intensity

$$\lambda_t = \mu_0 + \sum_{T_n^* < t} \alpha e^{-\eta(t-T_n^*)}, \quad \mu_0, \alpha, \eta > 0, \quad \alpha/\eta < 1.$$

We model the decision-time delay τ_n as exponential with rate b/λ_{T_n} for some baseline $b > 0$.

The Hawkes modeling has the following effect:

- A latency spike pushes λ up, the rate b/λ falls, and the agent waits longer between decisions.
- Hawkes self-excitation makes this slow regime persist after each spike — clusters of latency spikes propagate into clusters of long inter-decision waits.

In the locally Markovian approximation, specialising (3) with rate b/λ_{T_n} yields a state-dependent Laplace kernel,

$$K(s' | s_{T_n}, \lambda_{T_n}) = \frac{1}{2\beta_{\lambda_{T_n}}} e^{-|s'-s|/\beta_{\lambda_{T_n}}}, \quad \beta_\lambda := \sigma \sqrt{\frac{\lambda}{2b}}. \quad (6)$$

We can then describe the qualitative effect as follows:

- The kernel scale β_λ *grows* with λ : smoothing widens during clustered latency spikes and narrows in calm periods.
- the agent sees a wider effective reward landscape exactly when latency spikes cluster.

The Hawkes intensity is a learnable parameterization to augment the state-coordinates.

5 Latency as a landscape regularization

5.1 Effective reward and the policy shift

Theorem 3.1 converts the lagged problem on the baseline reward $r(a, s_{(t+\tau)-})$ — which depends on a *future* state at delivery time — into a problem with the *effective undelayed reward*

$$\tilde{r}(a, s) = \int_S r(a, s') \mathbb{E}_{\tau|s,a} [f_\tau(s' | s; a)] ds',$$

a function only of the action and the current state, on the original state space. The two formulations agree on every policy-evaluation functional, so the optimal policy under baseline r in the lagged dynamics coincides with the optimal policy under effective $\tilde{r}$ in the no-delay dynamics. The latency law μ enters only through the inter-decision transition kernel $\mathbb{E}_\tau[f_\tau]$: changing μ moves the kernel, which moves $\tilde{r}$.

Latency thus acts as a *physical* smoothing kernel on the reward landscape, with bandwidth set by μ rather than chosen as a regularization hyperparameter. As the mean delay increases, the kernel widens and $\tilde{r}$ smooths r on a longer scale — in the Gaussian-diffusion regime of §4.1 the effective bandwidth grows as $\sigma\sqrt{\tau_0}$. The scoring rule r is unchanged across latencies; what shifts is what the agent is effectively optimising.

The shape of the predicted policy shift is determined by which features of r survive convolution. A narrow peak of r — a “hit-or-miss” optimum with high mean reward but vanishing margin under execution noise — loses amplitude rapidly under widening of the kernel. A broad plateau loses amplitude more slowly. As μ widens, the global maximum of $\tilde{r}$ migrates from sharp peaks toward broader ones, and the corresponding optimal policy shifts from frame-perfect targeting to a more conservative policy that absorbs perturbations rather than threading them. §5.2 makes this concrete on a minimal one-dimensional environment in which the bifurcation between competing optima can be predicted in closed form.

5.2 A one-dimensional illustration

The agent steers a one-dimensional bar on $[0, W]$, $W = 40$, by issuing a discrete setpoint $a_t \in \{0, 1, \dots, 40\}$ at each step ($K = 41$ bins). The action is delivered after ε steps of latency, during which the bar drifts under independent Gaussian increments:

$$\text{bar}_t = a_{t-\varepsilon} + \sum_{s=t-\varepsilon+1}^t \xi_s, \quad \xi_s \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2),$$

with $\sigma = 0.6$. The conditional kernel $\text{bar}_t | a_{t-\varepsilon} \sim \mathcal{N}(a_{t-\varepsilon}, \sigma^2 \varepsilon)$ is exactly the smoothing kernel of §3, with variance linear in ε and collapsing to a Dirac at $\varepsilon = 0$.

A shot lands every $T = 10$ steps and rewards the bar’s instantaneous position against

$$r(x) = A_1 e^{-(x-\mu_1)^2/(2\sigma_1^2)} + A_2 e^{-(x-\mu_2)^2/(2\sigma_2^2)}, \quad (7)$$

with $(A_1, \mu_1, \sigma_1) = (2.5, 10, 0.6)$ a tall narrow peak and $(A_2, \mu_2, \sigma_2) = (1.0, 30, 5.0)$ a short wide plateau. Episodes are 500 steps; reward is zero between shots. The closed-form effective reward is itself a sum of Gaussians,

$$\tilde{r}(a; v) = \sum_{i \in \{1,2\}} A_i \cdot \frac{\sigma_i}{\sqrt{\sigma_i^2 + v}} e^{-(a-\mu_i)^2/(2(\sigma_i^2+v))}, \quad v = \sigma^2 \varepsilon, \quad (8)$$

which is what the agent receives in the smoothed-reward training mode of Theorem 3.1.

The amplitude of peak i in $\tilde{r}(\cdot; v)$ shrinks as $\tilde{A}_i(v) = A_i \sigma_i / \sqrt{\sigma_i^2 + v}$: narrow peaks (σ_i small) decay rapidly in v , wide peaks (σ_i large) decay slowly. As $v \rightarrow \infty$, $\tilde{A}_i(v) \sim A_i \sigma_i / \sqrt{v}$, so under enough smoothing the peaks rank by the product $A_i \sigma_i$ rather than by height A_i alone. With our parameters $A_1 \sigma_1 = 1.5$ and $A_2 \sigma_2 = 5.0$: the wide peak carries more than three times the asymptotic mass of the narrow peak even though the narrow peak is two and a half times taller. The argmax of $\tilde{r}(\cdot; v)$ therefore undergoes a discontinuous transition at the variance v_c where $\tilde{A}_1(v_c) = \tilde{A}_2(v_c)$,

$$v_c = \frac{(A_2 \sigma_2)^2 \sigma_1^2 - (A_1 \sigma_1)^2 \sigma_2^2}{(A_1 \sigma_1)^2 - (A_2 \sigma_2)^2}, \quad \varepsilon_c = v_c / \sigma^2. \quad (9)$$

For our parameters $v_c \approx 2.077$ and $\varepsilon_c \approx 5.77$.

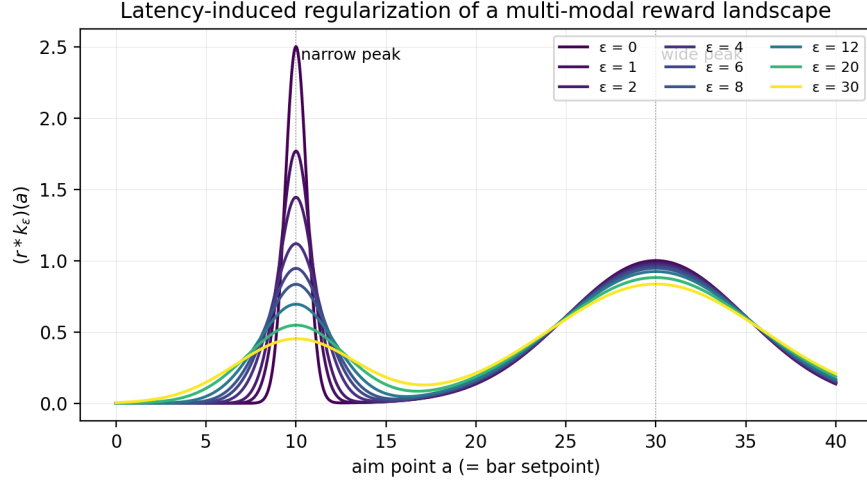


Figure 1: Effective reward landscape $\tilde{r}(a; v)$ for $\varepsilon \in \{0, 1, 2, 4, 6, 8, 12, 20, 30\}$.

Remark 5.1 (Latency as a Tikhonov-shaped robustness pressure). *The shrinkage law $\tilde{A}_i(v) = A_i \sigma_i / \sqrt{\sigma_i^2 + v}$ has the structural form of Tikhonov / ridge regularization: a quadratic penalty on the bandwidth-normalised amplitude, parameterised by $v = \sigma^2 \varepsilon$. The regularizer’s strength is not a free hyperparameter chosen for stability or exploration, as in entropy-regularized RL [2], but is uniquely determined by the inter-decision kernel of the delay (cf. Theorem 3.1). The bias toward broad optima is also structurally distinct from sharpness-aware minimization and the flat-minima literature [3, 4, 1]: there a smoothing kernel is imposed by the optimiser to bias toward generalisable parameters; here the kernel is physical, and the bias serves robustness to execution latency rather than train–test generalisation. The smoothed reward is the expected value of r under perturbed execution; an optimum that survives the perturbation is by construction one that does not require precise execution to be valuable.*

References

- [1] Pierre Foret et al. “Sharpness-aware minimization for efficiently improving generalization”. In: *International Conference on Learning Representations (ICLR)*. 2021.
- [2] M. Geist, B. Scherrer, and O. Pietquin. “A theory of regularized Markov decision processes”. In: (2019).
- [3] Sepp Hochreiter and Jürgen Schmidhuber. “Flat minima”. In: *Neural Computation* 9.1 (1997), pp. 1–42.
- [4] Nitish Shirish Keskar et al. “On large-batch training for deep learning: Generalization gap and sharp minima”. In: *International Conference on Learning Representations (ICLR)*. 2017.
- [5] M. Robin. “Contrôle impulsif avec retard pour les processus de diffusion”. In: *C. R. Acad. Sci. Paris Sér. A* 282 (1976), pp. 463–466.
- [6] M. Robin. “Contrôle impulsif des processus de Markov”. HAL: tel-00735779. Thèse de doctorat d’État ès sciences mathématiques. Université Paris IX-Dauphine, 1978.